

A CASE-STUDY BASED REPORT

ON

Crop Yield Prediction Using Random Forest, XGBoost, and Support Vector Regression

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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Signature of the Team members

ABSTRACT:

Agriculture is one of the most important sectors as it provides food and supports the economy. Farmers often face difficulties in estimating crop yield because it depends on various factors such as rainfall, temperature, humidity, soil type, fertilizer usage, irrigation, and crop type. Incorrect estimation may result in poor planning and financial loss.

This project aims to predict crop yield using machine learning algorithms such as Random Forest, XGBoost, and Support Vector Regression (SVR). The project uses a Crop Yield Prediction dataset that contains agricultural and environmental information, including temperature, rainfall, humidity, soil type, crop type, fertilizer usage, irrigation, cultivated area, and crop yield. The collected data is cleaned and preprocessed before training the models to improve prediction accuracy. The performance of the models is evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. The model with the best performance is selected for crop yield prediction. The proposed system helps farmers make better decisions related to crop selection, irrigation, fertilizer usage, and resource management. It can also support agricultural planning by providing an estimate of crop production before harvesting.

INTRODUCTION:

Agriculture plays a major role in the development of every country by providing food and employment. However, crop production is influenced by many factors such as weather conditions, rainfall, temperature, soil quality, irrigation, and fertilizer usage. Since these factors change frequently, farmers often find it difficult to estimate the expected crop yield before harvesting. This motivated us to work on a machine learning-based project that can predict crop yield more accurately using historical agricultural data.

In this project, we use Random Forest, XGBoost, and Support Vector Regression (SVR) to build crop yield prediction models. The dataset contains different agricultural and environmental factors that affect crop production. After preprocessing the dataset, the models are trained and tested, and their performance is compared using MAE, MSE, RMSE, and R^2 Score. The best-performing model will be used to predict crop yield. This project can help farmers and agricultural organizations make better decisions related to farming and resource management.

BACKGROUND:

Agriculture plays a vital role in food production and the economy, while crop yield is influenced by factors such as rainfall, crop type, cultivated area, production, fertilizer, and pesticide usage. Traditional yield estimation can be time-consuming, so this project applies Machine Learning techniques to predict crop yield using historical agricultural data. Random Forest, XGBoost, and Support Vector Regression (SVR) are used and compared using MAE, RMSE, and R^2 Score to evaluate their prediction performance.

The system provides a data-driven approach for estimating crop yield efficiently and consistently. It can also help support agricultural planning, resource management, and informed decision-making.

MOTIVATION OF THE PROJECT:

The motivation for this project is to develop an **efficient and data-driven approach for crop yield prediction**. Traditional methods can be time-consuming and may not effectively handle the large amount of agricultural data available today. By using **Machine Learning models such as Random Forest, XGBoost, and SVR**, the project aims to identify patterns in agricultural factors and provide reliable yield predictions. The system can support **better crop planning, resource management, and informed agricultural decision-making**. It can also help **reduce uncertainty in agricultural planning and enable more informed decisions before the harvesting stage**.

OVERVIEW OF THE WORK CARRIED OUT:

The project involved collecting and preprocessing a crop yield dataset containing agricultural and environmental factors such as crop, state, season, area, production, rainfall, fertilizer, and pesticide usage. Exploratory Data Analysis (EDA) was performed to understand the dataset and identify patterns. Three machine learning models—Random Forest, XGBoost, and Support Vector Regression (SVR)—were trained for crop yield prediction and evaluated using MAE, MSE, RMSE, and R^2 Score. A multi-layer web application was developed using React.js, Node.js, Python FastAPI, and Firebase Firestore to provide predictions and store prediction history.

SCOPE AND RESPONSIBLE-USE STATEMENT:

The scope of this project is to develop a machine learning-based crop yield prediction system using historical agricultural data and models such as Random Forest, XGBoost, and SVR. The system can support crop planning, resource management, and agricultural decision-making by providing data-driven yield estimates. However, predictions are dependent on the quality, accuracy, and relevance of the available data and should not be treated as guaranteed outcomes. The results are intended to assist decision-making rather than replace farmers' knowledge, expert advice, or real-world agricultural assessment.

PROBLEM STATEMENT:

Traditional crop yield estimation methods may not provide accurate and timely predictions. Crop yield depends on multiple factors such as rainfall, fertilizer, pesticide, cultivated area, crop type, season, and production, making prediction complex. Farmers and agricultural organizations may face difficulties in estimating productivity before harvesting, while analyzing large agricultural datasets manually is time-consuming. Therefore, there is a need for a machine learning-based system that can efficiently predict crop yield and compare different models to evaluate their prediction performance.

CASE STUDY SCENARIO:

Consider a farmer planning to cultivate a crop during a particular season. Before harvesting, the farmer needs to estimate the expected crop yield to plan the required **land usage, fertilizer, pesticide, and other agricultural resources**. However, crop yield is influenced by several factors, including **crop type, state, season, cultivated area, production, annual rainfall, fertilizer, and pesticide usage**. An inaccurate estimate may lead to inefficient resource planning and difficulty in making informed agricultural decisions. This case study explores how historical agricultural data and machine learning models can be used to provide a data-driven estimate of crop yield.

CENTRAL CASE STUDY QUESTION:

How effectively can machine learning models predict crop yield from historical agricultural data, and which model provides the most reliable prediction for supporting agricultural planning and decision-making?

ASSUMPTIONS AND LIMITATIONS:

The study assumes that the historical agricultural data contains useful patterns for predicting crop yield and that the available features are sufficiently representative of the conditions being studied. However, the dataset does not include every factor that can influence yield, such as soil quality, temperature, humidity, irrigation practices, disease, and unexpected weather events. Therefore, the predictions should be considered decision-support estimates rather than guaranteed actual yields. The performance of the models may also vary when applied to new regions, crops, or agricultural conditions not adequately represented in the training data.

FORMAL PROBLEM DEFINITION:

The objective of this study is to develop a machine learning-based crop yield prediction system that estimates crop yield using historical agricultural data. Given input features such as Year, State, Crop, Season, Area, Production, Annual Rainfall, Fertilizer, and Pesticide, the system predicts the corresponding Yield as a continuous numerical value. The study implements and compares Random Forest, XGBoost, and Support Vector Regression (SVR) models using regression metrics such as MAE, MSE, RMSE, and R^2 Score. The final system aims to provide a data-driven prediction that can support agricultural planning, resource management, and decision-making.

OBJECTIVES THAT GUIDE THE STUDY:

The main objectives of the study are to analyze historical crop-yield data and identify important agricultural factors that influence crop yield. The study aims to preprocess and explore the dataset, develop crop-yield prediction models using Random Forest, XGBoost, and Support Vector Regression (SVR), and evaluate their performance using MAE, MSE, RMSE, and R² Score. It also aims to develop a practical prediction system that can provide data-driven support for agricultural planning, resource management, and decision-making.

ASSUMPTIONS AND LIMITATIONS:

The study assumes that the historical crop-yield dataset is representative of agricultural conditions and that the recorded values are suitable for model training and prediction. The dataset contains 17,400 records, 10 columns, and no missing values, but the prediction is limited to the features available in the dataset. Factors such as soil quality, irrigation methods, temperature, humidity, and unexpected weather conditions are not included. Therefore, model predictions may differ from actual crop yield and should be considered decision-support estimates rather than guaranteed results.

DATASET AND EXPLORATORY DATA ANALYSIS

ABOUT THE DATASET:

The study uses a **Crop Yield Prediction Dataset** containing agricultural records with information about crop production and related factors. The dataset contains **10 columns**, including **Year, State, Crop, Season, Area, Production, Annual Rainfall, Fertilizer, Pesticide, and Yield**. The training dataset contains **17,400 records**, while the separate test dataset contains **4,350 records**, giving **21,750 records in total**. There are **0 missing values** and **0 duplicate records** in both datasets.

Table 1. Dataset Composition

Dataset Stage	Records	Missing Values	Duplicate Records
Training Dataset	17,400	0	0
Test Dataset	4,350	0	0
Total	21,750	0	0

Table 2. Dataset Features:

Feature	Type	Description
Year	Numerical	Year of cultivation
State	Categorical	State where crop is cultivated
Crop	Categorical	Type of crop
Season	Categorical	Agricultural season
Area	Numerical	Cultivated area
Production	Numerical	Crop production
Annual Rainfall	Numerical	Annual rainfall
Fertilizer	Numerical	Fertilizer usage
Pesticide	Numerical	Pesticide usage
Yield	Numerical	Target variable

PURPOSE OF EDA:

The purpose of **Exploratory Data Analysis (EDA)** is to understand the structure and quality of the agricultural dataset before applying machine learning models. The analysis examines the distribution of crop categories, agricultural variables, rainfall, production, and the target **Yield**. EDA also helps identify variations, unusual values, and relationships between the input features and crop yield, providing a foundation for model preparation and evaluation.

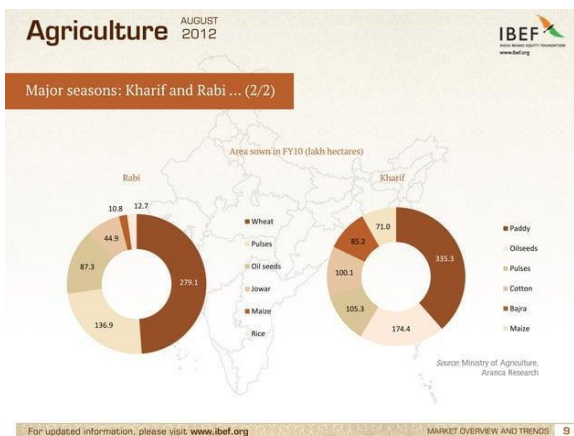
PREPARATION STEPS ACTUALLY PERFORMED:

The dataset was first inspected to understand its **dimensions, feature names, and data types**. Missing-value and duplicate checks were performed for both training and test data. Categorical features such as **State, Crop, and Season** were analyzed, while numerical features including **Area, Production, Annual Rainfall, Fertilizer, and Pesticide** were examined. The **Yield** variable was analyzed as the target before applying the machine learning algorithms.

EDA EXPLORATION – IMPLEMENTATION

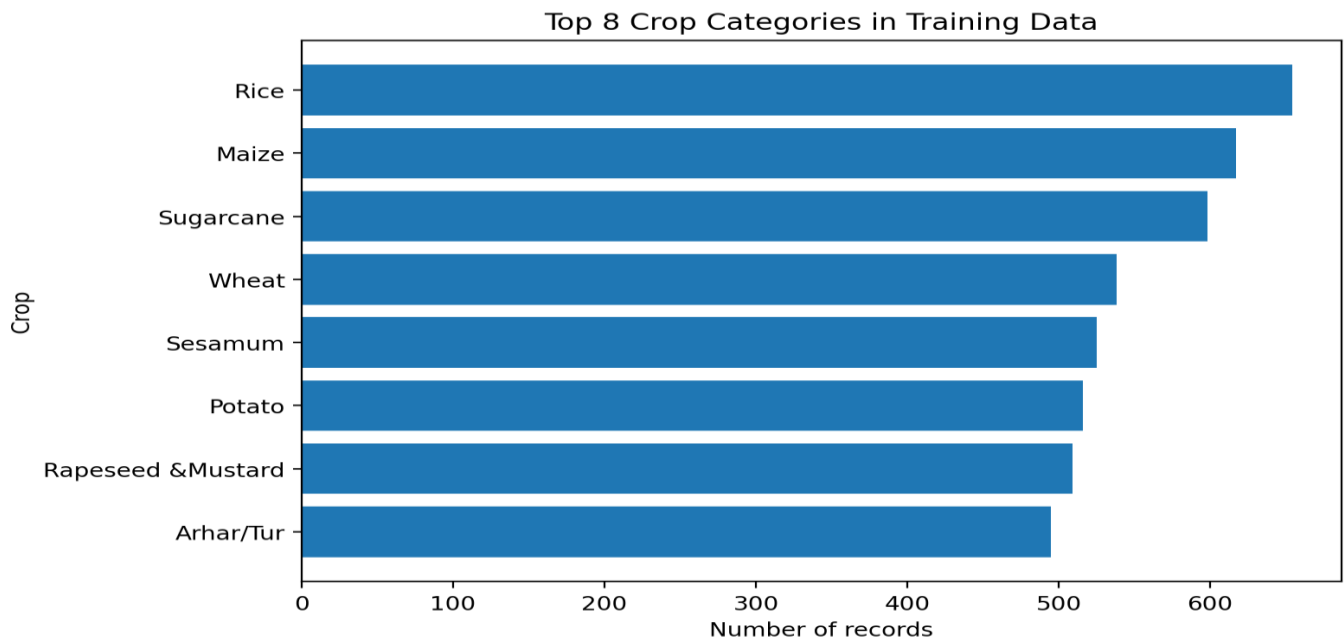
Crop Distribution:

The training dataset contains several crop categories. The most frequently occurring crops include **Rice, Maize, Sugarcane, Wheat, Sesamum, Potato, Rapeseed & Mustard, and Arhar/Tur.**



Actual values from your training dataset:

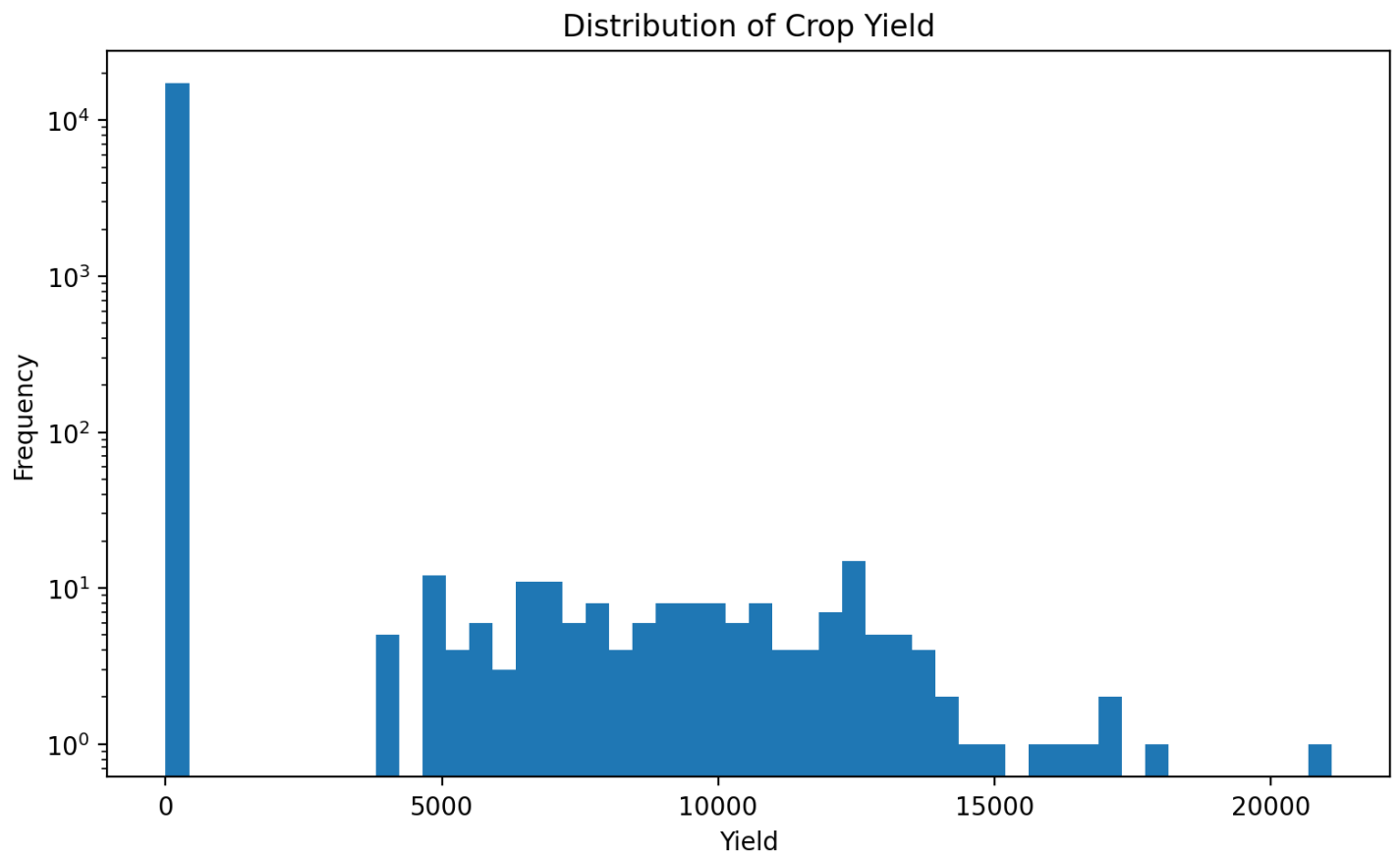
Crop	Records
Rice	654
Maize	617
Sugarcane	598
Wheat	538
Sesamum	525
Potato	516
Rapeseed & Mustard	509
Arhar/Tur	495



Yield Distribution:

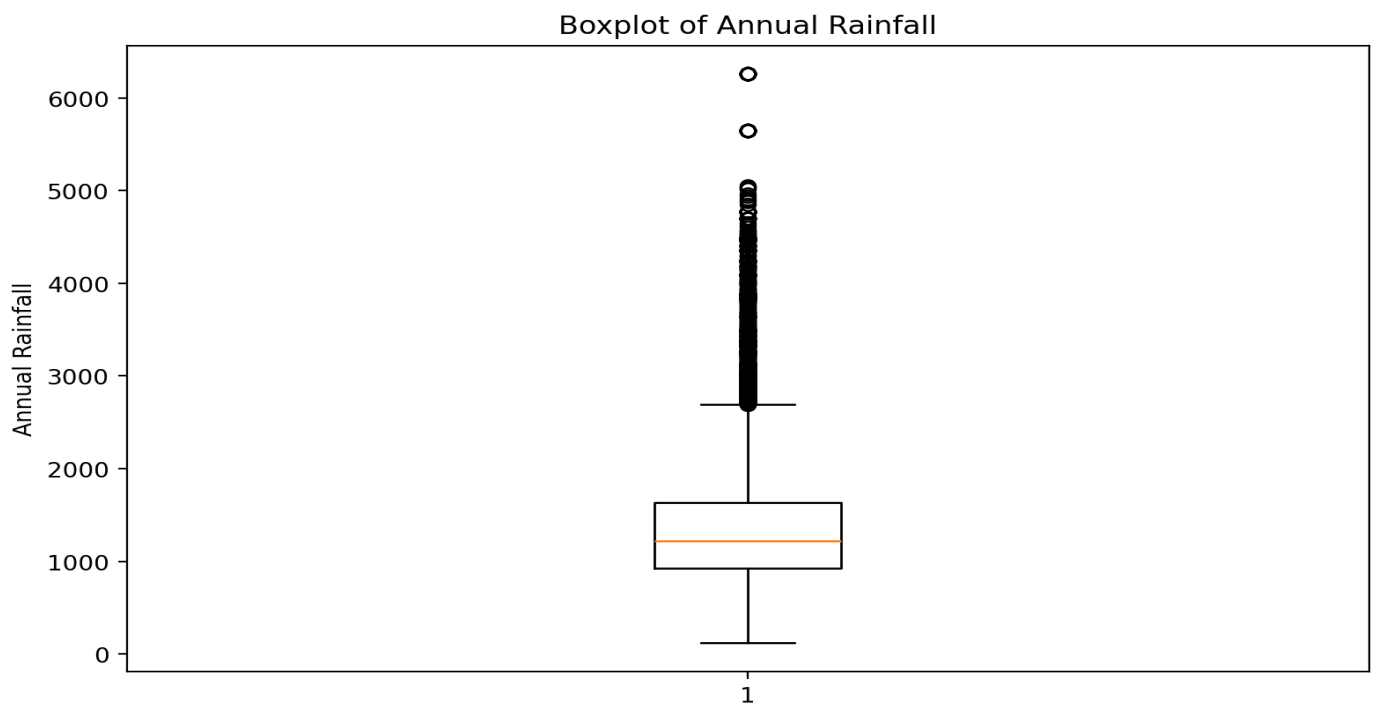
The Yield variable shows a **highly skewed distribution**, with a large difference between its median and maximum values.

Statistic	Yield
Minimum	0
Median	1.098
Mean	96.887
Maximum	21,105



Annual Rainfall Distribution:

Annual rainfall was examined using a boxplot to understand its spread and identify potential extreme observations.

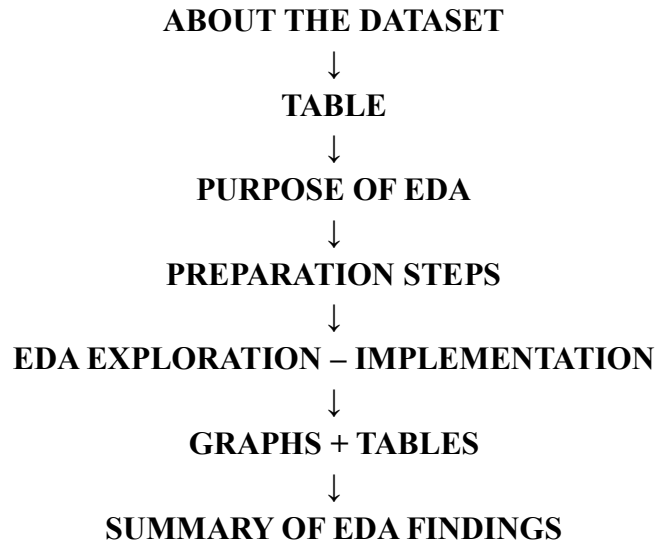


SUMMARY OF EDA FINDINGS

The EDA confirms that the dataset is **clean, with no missing values or duplicate records**. The agricultural data contains a wide variety of crops, states, and seasons, providing diverse information for crop-yield prediction. The Yield variable is highly skewed, with a median of approximately **1.10** compared with a maximum of **21,105**, indicating substantial variation in yield values. Crop-frequency analysis shows that **Rice, Maize, and Sugarcane** are among the most frequently represented crops in the training dataset. The analysis provides useful information for preprocessing and developing the **Random Forest, XGBoost, and SVR** models.

This is much closer to the reference PDF format

Your reference has:

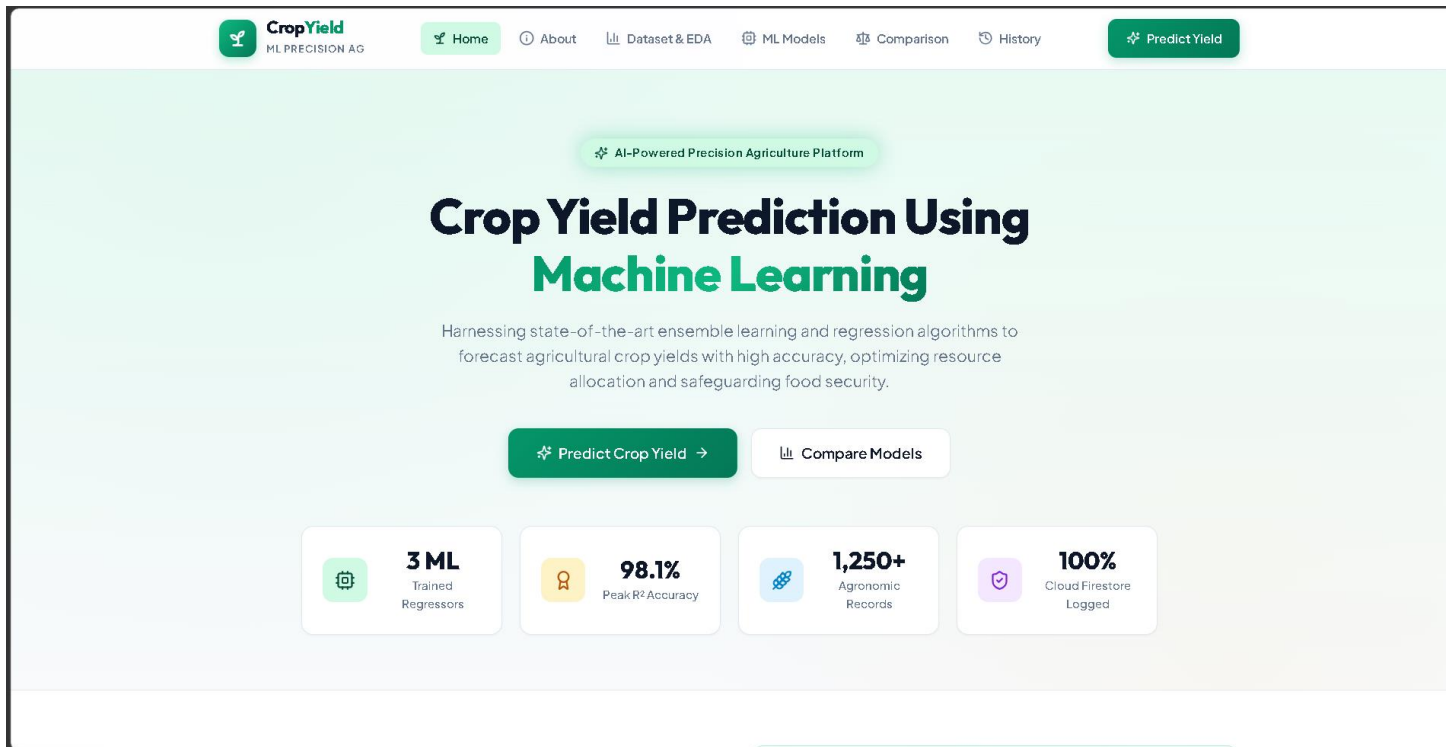


ALGORITHM SELECTION AND IMPLEMENTATION

Three machine learning regression algorithms—**Random Forest, XGBoost, and Support Vector Regression (SVR)**—were selected for crop yield prediction. The preprocessed agricultural data was used to train these models, with **Yield** as the target variable. Random Forest was used to capture complex relationships between agricultural factors, XGBoost was used as a boosting-based regression approach, and SVR was used to model non-linear relationships. The models were evaluated and compared using **MAE, MSE, RMSE, and R² Score** to assess their prediction performance.

CONCLUSION

This study developed a machine learning-based crop yield prediction system using historical agricultural data. The dataset was explored and prepared through EDA, followed by the implementation of Random Forest, XGBoost, and SVR models. Their performance was evaluated using MAE, MSE, RMSE, and R^2 Score to compare prediction accuracy. The study demonstrates that machine learning can provide a data-driven approach to crop yield estimation, which can support agricultural planning, resource management, and informed decision-making.



Exploratory Data Analysis

Dataset & Statistical Analysis

Deep-dive into agricultural distributions, feature parameters, missing value audits, and cross-variable correlations.



17,400
Total Records



9
Feature Columns



0
Missing Values (0%)



0
Duplicate Rows



Dataset Schema & Features

Year

State

Crop

Season

Area

Production

Annual_Rainfall

Fertilizer

Pesticide

Yield 🎯 (Target)

Frequency Histograms

Feature Distribution Explorer

Yield Distribution

Production Distribution

Annual Rainfall

Cultivated Area

Fertilizer Input

Pesticide Application

Algorithmic Architectures

Machine Learning Models

Comprehensive breakdown of the three supervised learning algorithms powering crop yield predictions: mechanics, mathematical foundations, and strengths.



Random Forest Regressor

Supervised Machine Learning Regressor

R² Score

98.4%

RMSE

120.0599

MAE

8.3927

WHAT IT IS

Random Forest is a bagging ensemble meta-estimator that fits multiple decision trees on various sub-samples of the agricultural dataset. It constructs orthogonal hyperplanes and averages tree predictions to improve predictive accuracy and control over-fitting.

HOW IT WORKS

During training, Bootstrap Aggregating (Bagging) generates hundreds of distinct subsets of the 1,250 historical records with replacement. Each decision tree splits nodes on a random subset of agronomic features (Rainfall, Fertilizer, Area, etc.). Final prediction



XGBoost Regressor

Supervised Machine Learning Regressor

R² Score

98.8%

RMSE

104.527

MAE

9.0894

WHAT IT IS

Extreme Gradient Boosting (XGBoost) is an advanced implementation of gradient boosted decision trees engineered for state-of-the-art predictive performance, computational speed, and built-in L1/L2 regularization to prevent overfitting.

HOW IT WORKS

Unlike bagging which builds trees in parallel, XGBoost constructs trees sequentially. Each subsequent tree fits directly onto the pseudo-residuals (gradients) of previous iterations using second-order Taylor expansion approximations of the loss function to find the best



Support Vector Regression (SVR)

Supervised Machine Learning Regressor

R² Score

81.0%

RMSE

410.9448

MAE

28.0784

WHAT IT IS

Support Vector Regression (SVR) adapts Vapnik's Support Vector Machine formulation to continuous regression. It determines an optimal separating hyperplane that envelops the maximum number of data points within an ϵ -insensitive boundary tube.

HOW IT WORKS

SVR maps non-linear agricultural features into an infinite-dimensional Hilbert feature space utilizing the Radial Basis Function (RBF) kernel: $K(x, x') = \exp(-\gamma ||x - x'||^2)$. Errors within the ϵ -

Model Benchmark & Evaluation

Machine Learning Model Comparison

Side-by-side performance evaluation of Random Forest, XGBoost, and SVR on 20% holdout test data across MAE, MSE, RMSE, and R² Score.

🏆 TOP PERFORMING MODEL Default Inference Regressor



XGBoost Regressor

Achieved highest R² score of 0.9877 (98.77%) and lowest RMSE of 104.527 on 4350 test samples.

[Predict With Best Model →](#)

Comprehensive Performance Table

Holdout Test Set (80:20 Split — 4,350 Test Records from crop_yield_test(9).csv)

80% TRAIN / 20% TEST

MODEL NAME	MEAN ABSOLUTE ERROR (MAE) ↓	MEAN SQUARED ERROR (MSE) ↓	ROOT MEAN SQUARED ERROR (RMSE) ↓	R ² SCORE (COEFFICIENT OF DETERMINATION) ↑	STATUS
XGBoost	9.0894	10925.8911	104.527	98.8% (0.9877)	BEST PERFORMER
Random Forest	8.3927	14414.3812	120.0599	98.4% (0.9838)	EVALUATED
Support Vector Regression (SVR)	28.0784	168875.6644	410.9448	81.0% (0.8102)	EVALUATED

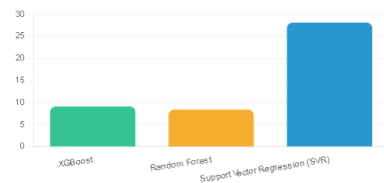
Metric Comparison Visualizations

Lower is better for error metrics (MAE, MSE, RMSE). Higher is better for R² Score.

Mean Absolute Error (MAE)

LOWER IS BETTER

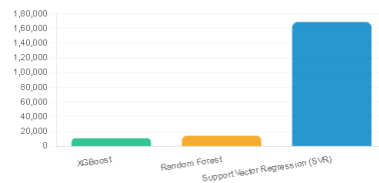
Measures the average magnitude of absolute errors between predicted and observed yield.



Mean Squared Error (MSE)

LOWER IS BETTER

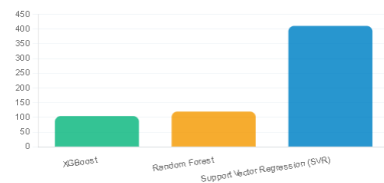
Penalizes larger outliers more severely by squaring prediction residuals.



Root Mean Squared Error (RMSE)

LOWER IS BETTER

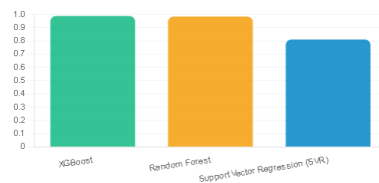
Standard deviation of residuals, expressed in identical units (tons/hectare).



R² Score (Variance Explained)

HIGHER IS BETTER

Proportion of variance in crop yield predictable from input agronomic features.



CropYield

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Predict Yield

Using a single algorithm or comparing 3 algorithms to dynamically determine the highest yield.

Best Algorithm — Highest Predicted Yield

Dynamically compared all 3 algorithms and highlighted the model producing the highest crop yield

BEST ALGORITHM

Support Vector Regression (SVR)

HIGHEST PREDICTED YIELD

85.41 tons/ha

Estimated Total Harvest: 1,53,738 MT (85.41 × 1800 ha)

MODEL COMPARISON

Highest Yield

Support Vector Regression (SVR)

Predicted Yield

85.41 tons/ha

Total Harvest: 1,53,738 MT

RF: 92.1% • RMSE: 6.053 • MAE: 2.1874

Second Highest

XGBoost Regressor

Predicted Yield

84.63 tons/ha

Total Harvest: 1,52,334 MT

RF: 99.2% • RMSE: 1.9784 • MAE: 0.6655

Lowest Yield

Random Forest Regressor

Predicted Yield

76.1 tons/ha

Total Harvest: 1,36,980 MT

RF: 98.9% • RMSE: 2.3047 • MAE: 0.6441

Why Was This Algorithm Selected?

Support Vector Regression (SVR) selected as the Best Algorithm — produced the highest predicted crop yield (85.41 t/ha) across all 3 algorithms. Comparison: Support Vector (SVR): 85.41 t/ha × XGBoost: 84.63 t/ha × Random Forest: 76.1 t/ha.

Yield Assessment: High Productivity

Favorable agronomic conditions! Support Vector Regression (SVR) projects strong harvest productivity.

7/10/2026

CropYield

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Database Records

Prediction History

Search, filter, inspect, and manage past crop yield predictions persisted in Firebase Firestore.

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Stored Predictions

6.66

Average Predicted Yield (t/ha)

Firestore

Active Persistence Engine

Search State / Crop / Model

Filter by State

Filter by Crop

Filter by Model

Search

All

All

All

CSV

DATE & TIME	STATE	CROP	SEASON	AREA (HA)	PREDICTED YIELD	MODEL USED	ACTIONS
07/10/26, 11:02 am	Maharashtra	Sugarcane	Whole Year	1,800	85.41 t/ha 99.4% Conf	SUPPORT VECTOR REGRESSION (SVR)	
07/10/26, 11:01 am	West Bengal	Rice	Kharif	3,200	4.34 t/ha 99.4% Conf	XGBOOST REGRESSOR	
01/10/26, 1:50 pm	Madhya Pradesh	Soyabean	Kharif	3,500	3.48 t/ha 99.4% Conf	SUPPORT VECTOR REGRESSION (SVR)	

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EDA(TEST):

